

Problem 1 – Secret Diary

```
memory = input("Enter today's memory: ")  
with open("diary.txt", "a") as file:  
    file.write(memory + "\n")  
print("\nAll Saved Memories:")  
with open("diary.txt", "r") as file:  
    print(file.read())
```

Problem 2 – Class Attendance

```
with open("attendance.txt", "r") as file:  
    students = file.readlines()  
print("Total Attendance:", len(students))
```

Problem 3 – Shopping History

```
with open("orders.txt", "r") as file:  
    orders = file.readlines()  
print("Orders:")  
for item in orders:  
    print(item.strip())  
print("Total Orders:", len(orders))
```

Problem 4 – Password Inspector

```
password = input("Enter Password: ")  
upper = 0  
lower = 0  
digits = 0  
for ch in password:  
    if ch.isupper():
```

```
        upper += 1
    elif ch.islower():
        lower += 1
    elif ch.isdigit():
        digits += 1

print("Uppercase:", upper)
print("Lowercase:", lower)
print("Digits:", digits)

if len(password) >= 8 and upper > 0 and lower > 0 and digits > 0:
    print("Strong Password")
else:
    print("Weak Password")
```

Problem 5 – Chat Message Analyzer

```
message = "Python is Fun and Python is Powerful"
print("Python Count:", message.count("Python"))
print("Uppercase:", message.upper())
print("Modified:", message.replace("Powerful", "Amazing"))
```

Problem 6 – Flight Ticket

```
city = input("Enter City Name: ")
city = city.strip().title()
print("Cleaned City:", city)
```

Problem 7 – ATM Withdrawal

```
def withdraw(balance, amount):
    if amount <= balance:
        return balance - amount
```

```
else:
    return "Insufficient Balance"
balance = float(input("Enter Balance: "))
amount = float(input("Enter Withdrawal Amount: "))
print("Result:", withdraw(balance, amount))
```

Problem 8 – Grade Calculator

```
def calculate_grade(mark):
    if mark >= 90:
        return "A"
    elif mark >= 75:
        return "B"
    elif mark >= 60:
        return "C"
    else:
        return "F"
mark = int(input("Enter Marks: "))
print("Grade:", calculate_grade(mark))
```

Problem 9 – Bill Calculator

```
def calculate_bill(price, quantity):
    return price * quantity
price = float(input("Enter Price: "))
quantity = int(input("Enter Quantity: "))
bill = calculate_bill(price, quantity)
print("Final Bill: ₹", bill)
```

Problem 10 – Discount Calculator

```
discount = lambda price: price - (price * 0.10)
```

```
price = float(input("Enter Product Price: "))
```

```
print("Price After Discount:", discount(price))
```

Problem 11 – Rectangle Area

```
area = lambda length, width: length * width
```

```
length = float(input("Enter Length: "))
```

```
width = float(input("Enter Width: "))
```

```
print("Area:", area(length, width))
```

Problem 12 – Speed Checker

```
speed_check = lambda speed: "Overspeed" if speed > 80 else "Normal Speed"
```

```
speed = int(input("Enter Speed: "))
```

```
print(speed_check(speed))
```